

Problem definition

Problem. Hate in video lives jointly in visuals, audio delivery, and speech. Unimodal classifiers miss cases where the three signals must combine to separate hate from satire or critical discourse.

Hypothesis. A small fusion transformer on frozen vision/audio/text encoders can beat every unimodal baseline, but the gain is not uniform across metrics or communities.

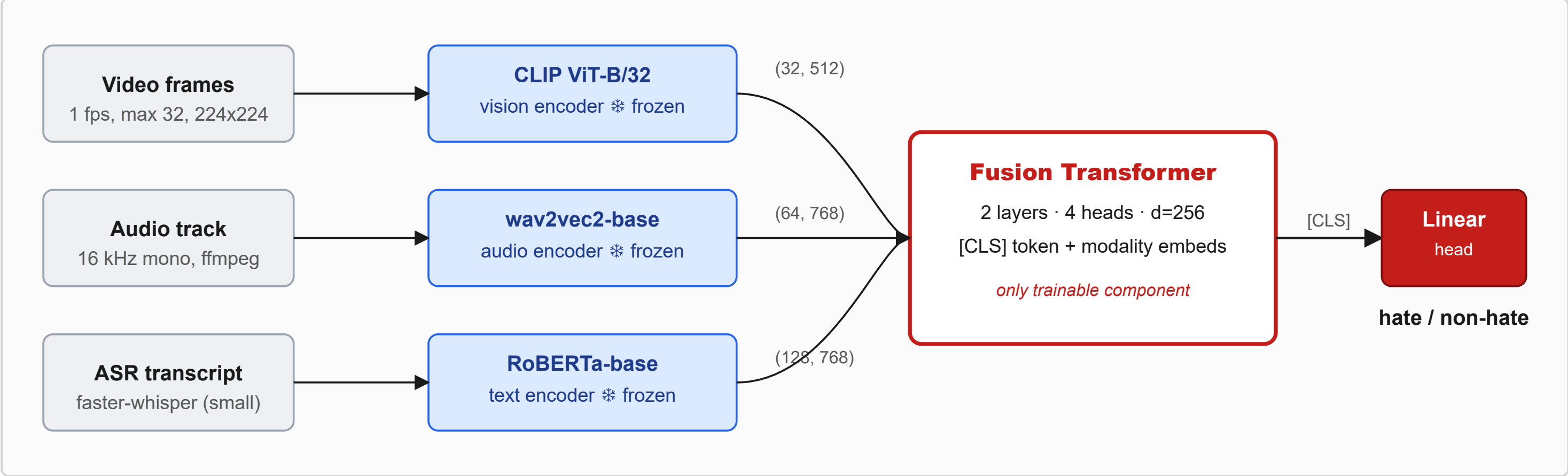
Dataset

- HateMM [1]: 1,083 BitChute videos** (431 hate, 652 non-hate); ~43.5 hours, 6.3 GB. CC BY 4.0.
- A video is labeled *hate* when it attacks or demeans someone for who they are: religion, race, ethnicity, gender, sexual orientation, or disability. Each hate video also tags the targeted community.
- 80/10/10 stratified split by binary label (863/109/109), fixed seed; preprocessing succeeded on 1081/1083 videos.
- Scope:** BitChute is a low-moderation platform, so our results characterise hate detection on that source distribution; transfer to YouTube or TikTok would require recalibration.

Method

- Preprocess.** Sample frames at 1 fps capped at 32 per video, center-cropped to 224 x 224; extract 16 kHz mono audio via ffmpeg; transcribe speech with faster-whisper.
- Frozen feature extraction.** CLIP ViT-B/32 per frame, wav2vec2-base mean-pooled into 64 audio tokens, RoBERTa-base token embeddings (length 128) on the transcript.
- Fusion.** All token sequences are concatenated, prepended with a [CLS] token and per-modality type embeddings, then fed into a 2-layer pre-norm transformer (d = 256, 4 heads, GELU). The [CLS] output goes to a binary linear classifier.

Why frozen encoders. Frozen encoders make the ablation fair: only the fusion head varies. Restricting it to one modality yields each unimodal baseline.



Related works

- HateMM [1].** Defines the task and releases the dataset and target-community labels we audit against; their best multimodal baseline reaches ~0.80 macro-F1.
- Hateful Memes [2].** Foundational multimodal hate benchmark (image + text); MOMENTA [3] adds context-aware fusion on the same line of work.
- Attention Bottlenecks (Nagrani et al.) [4].** Show that cross-modal attention can match heavy late-fusion pipelines at lower cost; we use a lighter variant suited to our dataset size.
- Large pretrained encoders [5, 6, 7]** transfer well to downstream tasks with only a small head trained on top; the frozen-encoder paradigm we rely on here.

The Seed Trap

Seed 42 alone showed fusion beating text by +5.5 accuracy points . Across three seeds {42, 1337, 2025} the gap vanished completely .				
Model	Seed 42 alone		3-seed mean ± std	
Fusion (V+A+T)	83.5% acc	F1 0.824	79.2% ± 5.2%	F1 0.782 ± 0.055
Text only	78.0% acc	F1 0.763	79.2% ± 2.1%	F1 0.781 ± 0.021
Gap	+5.5 pp		0.0 pp, dead tie	
Only AUROC survived (0.876 vs 0.827); a single seed would have shipped a false claim.				

Validation

Table 1. Modality ablation on the test split (n = 109), mean ± std over 3 seeds {42, 1337, 2025}. Our 0.78 macro-F1 is comparable to Das et al.'s reported ~0.80 on the same dataset.

Modality	Acc	F1	Prec	Rec	AUROC
V + A + T	0.79 ± 0.05	0.78 ± 0.05	0.74 ± 0.08	0.74 ± 0.08	0.88 ± 0.01
V + A	0.77 ± 0.01	0.76 ± 0.00	0.71 ± 0.04	0.71 ± 0.08	0.85 ± 0.01
V + T	0.78 ± 0.03	0.77 ± 0.02	0.73 ± 0.07	0.72 ± 0.04	0.86 ± 0.01
A + T	0.76 ± 0.02	0.75 ± 0.02	0.71 ± 0.05	0.69 ± 0.08	0.84 ± 0.00
Text only	0.79 ± 0.02	0.78 ± 0.02	0.75 ± 0.06	0.73 ± 0.08	0.83 ± 0.02
Video only	0.72 ± 0.03	0.71 ± 0.02	0.64 ± 0.06	0.68 ± 0.06	0.81 ± 0.01
Audio only	0.67 ± 0.02	0.67 ± 0.03	0.56 ± 0.02	0.71 ± 0.05	0.77 ± 0.03

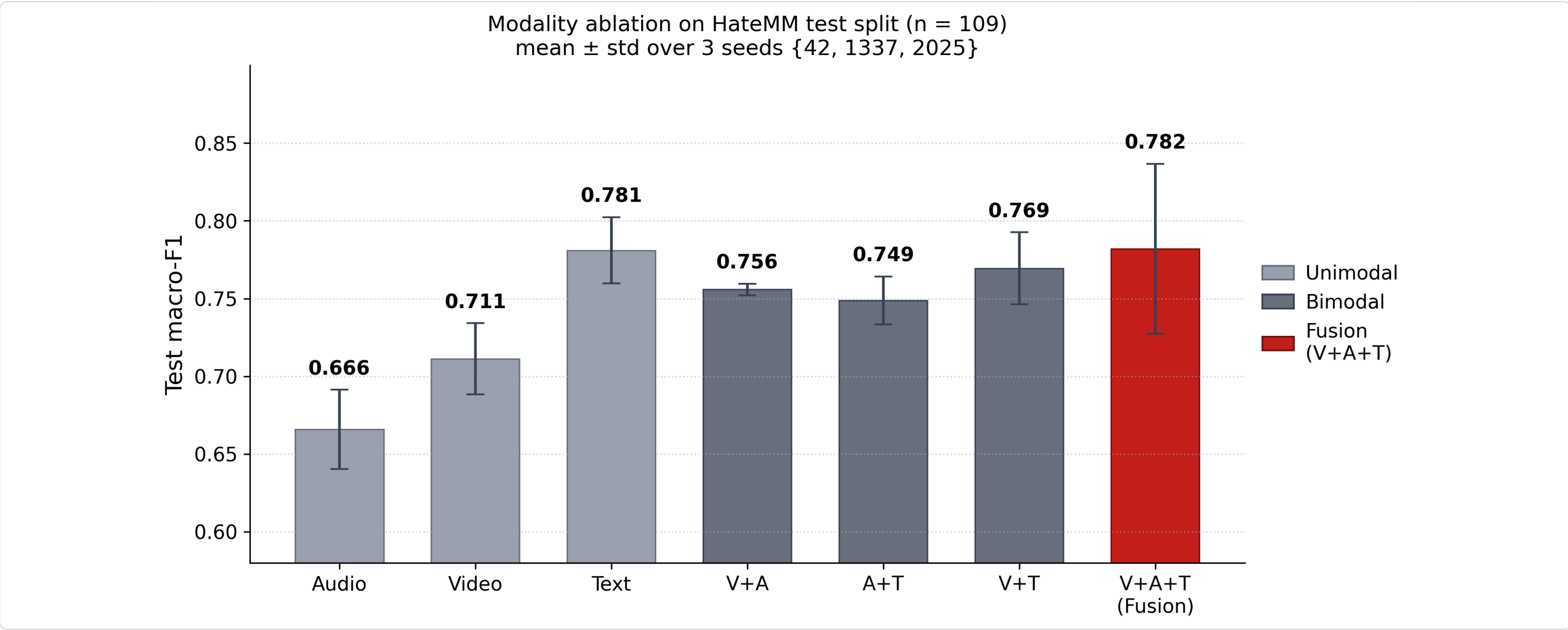


Figure 1. Test macro-F1 for all seven modality configurations, mean ± std over 3 seeds {42, 1337, 2025}.

Key observations

Each added modality lifts AUROC monotonically. From audio alone (0.77) to full fusion (0.88), every step in the seven-row ablation buys ranking signal. Accuracy, in contrast, plateaus at the text-only level (0.79).

Video is the stronger complement to text. Text → V+T adds +0.030 AUROC; text → A+T adds only +0.012. Adding the third modality on top of any pair gains another ~+0.02.

Fairness Analysis

Table 2. Per-community test accuracy (n ≥ 15), mean ± std over 3 seeds.

Community	n	Fusion	Video	Audio	Text
Blacks	46	0.75 ± 0.05	0.70 ± 0.04	0.59 ± 0.06	0.73 ± 0.03
Others	39	0.84 ± 0.01	0.74 ± 0.09	0.68 ± 0.11	0.85 ± 0.05
Jews	15	0.73 ± 0.13	0.67 ± 0.07	0.71 ± 0.04	0.76 ± 0.04

- ⚠ **Others (n = 39)** is predominantly non-hate in our test split; high accuracy reflects majority-class prediction, not genuine discrimination.
- Text wins on *Others* and *Jews*; fusion leads only on *Blacks*.
- No single model is best across all communities.

Limitations

- Small test set (n = 109).** Reported per-community stats (n ≥ 15) still carry appreciable variance.
- Platform specificity.** BitChute-trained models may not transfer to lower-toxicity platforms with different style.
- ASR quality varies.** Shouting, music, and accented speech degrade the text modality.
- Frozen backbones cap performance.** Audio mean-pools to 64 fixed slots, losing detail on long videos. Unfreezing would lift numbers but break the controlled ablation.

Conclusion

- AUROC improves robustly.** 0.88 ± 0.01 vs 0.83 ± 0.02 for text alone.
- Accuracy and F1 do not.** Across three seeds the gain disappears into seed variance.
- Interpretation.** Fusion *rank*s videos better; it does not classify more correctly.

Practical implication. Three deployment options:

- V + A + T (AUROC 0.88 ± 0.01):** when ranking quality matters most.
- V + T (AUROC 0.86 ± 0.01):** cost-aware sweet spot, within 0.02 AUROC of fusion without an audio pipeline.
- Text-only (acc 0.79 ± 0.02):** cheapest fixed-threshold option, ties V+A+T on accuracy.

References

1. M. Das et al., "HateMM," *ICWSM*, 2023.
2. D. Kiela et al., "Hateful Memes Challenge," *NeurIPS*, 2020.
3. S. Pramanick et al., "MOMENTA," *EMNLP Findings*, 2021.
4. A. Nagrani et al., "Attention Bottlenecks for Multimodal Fusion," *NeurIPS*, 2021.
5. A. Radford et al., "CLIP," *ICML*, 2021.

6. A. Baevski et al., "wav2vec 2.0," *NeurIPS*, 2020.
7. Y. Liu et al., "RoBERTa," *arXiv:1907.11692*, 2019.



Scan for code
github.com/YDIB11/Mini-Project
EE-559 Deep Learning · EPFL 2026